CHM111 Practice Exam 3 on chapter(s): 5, 6

Name:____

These questions assess your understanding of the material from Chapters 5, 6 of Openstax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

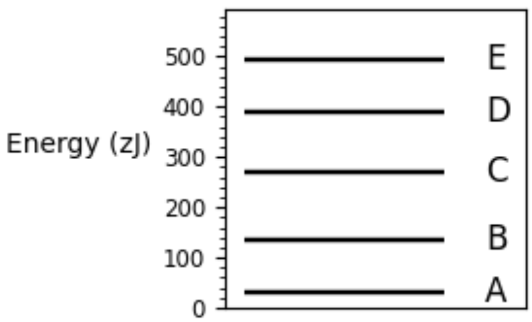
1. Measurements show that the enthalpy of a mixture of gaseous reactants has decreased by 185 kJ during a certain chemical reaction, which was carried out at a constant pressure. Evidenced by the volume change, -192 kJ of work must have been done on the mixture during the reaction. Calculate the change in energy of the gas mixture during the reaction and report whether is it endothermic, exothermic, or neither.

   ANSWER: ____________________________

2. A particular ear of corn has an energy content of 65.3 kcal. What is this in kilojoules?

   ANSWER: ____________________________

3. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the highest frequency?

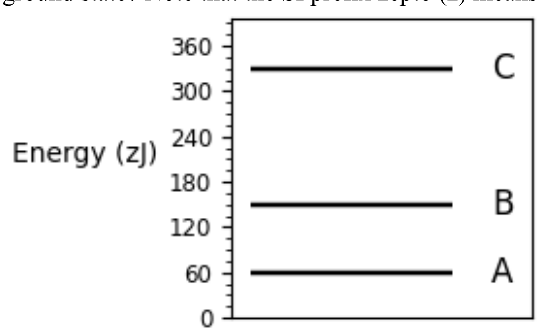


   ANSWER: ____________________________

4. It takes 153 kJ/mol to break a bond between two particular elements. Calculate the maximum wavelength of light that will break one of these bonds.

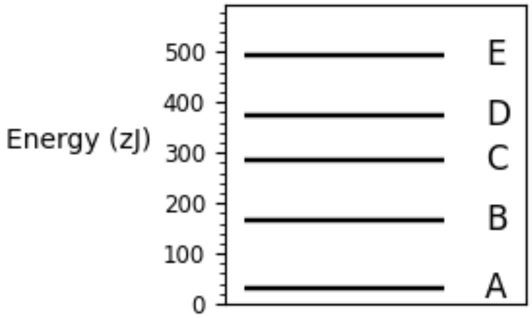
   ANSWER: ____________________________

5. This energy diagram shows the allowed energy levels of an electron in a certain atom. What is the energy of the electron in its ground state? Note that the SI prefix zepto (z) means 10^{-21}.



   ANSWER: ____________________________

6. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the longest wavelength?



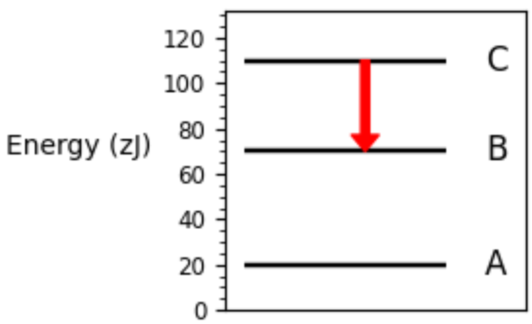
ANSWER: ______________________________

7. Fill in the information missing from the following table. For each subshell, you must provide the principal quantum number n, the angular momentum quantum number l, and the maximum number of electrons that can fit in the subshell N_e. Credit is awarded only if the table is completed correctly.

Subshell	n	l	N_e
1s			
2s			
2p			

ANSWER: ______________________________

8. This energy diagram shows the allowed energy levels of an electron in a certain atom. If the electron makes a transition from C to B, as shown by the arrow, will a photon be absorbed or emitted?

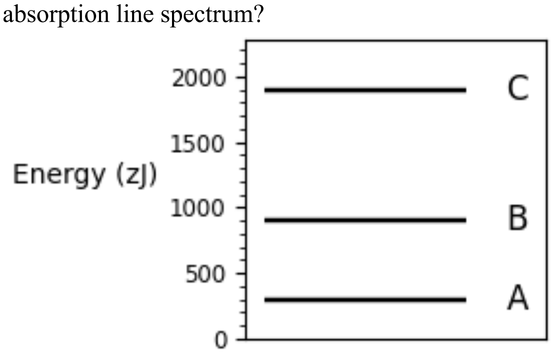


ANSWER: ______________________________

9. A mixture of dinitrogen monoxide and neon gas is expanded from a volume of 43.5 L to a volume of 62.2 L while the pressure is held constant at 54.7 atm. Calculate the work done on the gas mixture and report your answer to 3 significant figures.

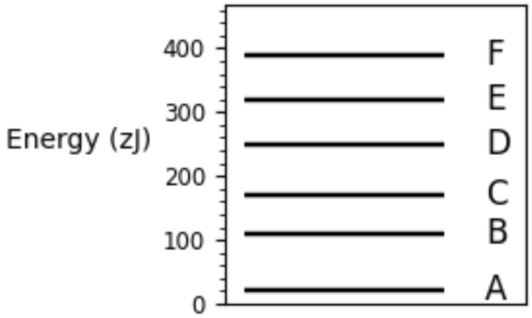
ANSWER: ______________________________

10. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. How many lines are in the absorption line spectrum?



ANSWER: ______________________________

11. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the shortest wavelength?



ANSWER: ____________________________

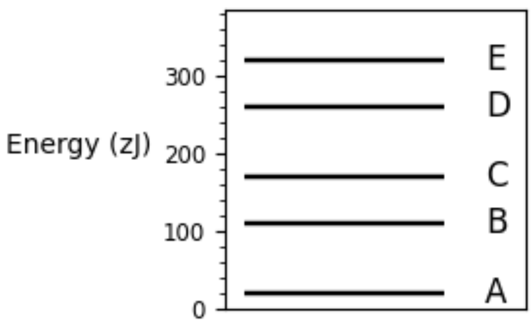
12. Several types of electromagnetic radiation (EMR) are: yellow light, red light, orange light, and ultraviolet light. Order these types of EMR from lowest frequency to highest frequency.

ANSWER: ____________________________

13. Calculate the wavelength of the line in the emission line spectrum of hydrogen caused by the transition of the electron from an orbital with $n = 11$ to an orbital with $n = 4$. Round your answer to 3 significant figures.

ANSWER: ____________________________

14. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. How many lines are in the emission line spectrum?

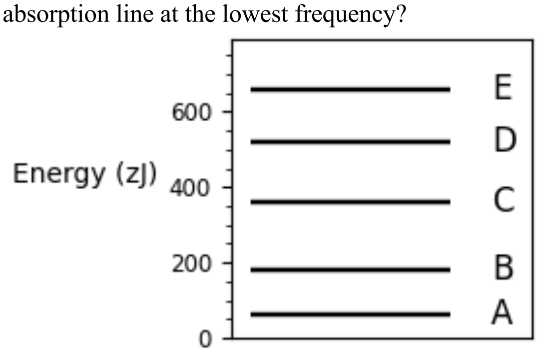


ANSWER: ____________________________

15. Several types of electromagnetic radiation (EMR) are: gamma rays, orange light, red light, and green light. Order these types of EMR from lowest energy to highest energy.

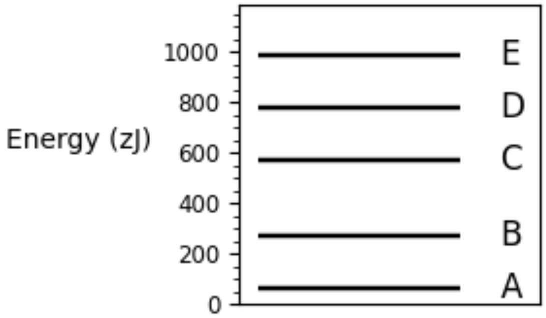
ANSWER: ____________________________

16. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the lowest frequency?



ANSWER: ____________________________

17. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the lowest frequency?



ANSWER: ____________________________

18. Several types of electromagnetic radiation (EMR) are: ultraviolet light, blue light, microwaves, and orange light. Order these types of EMR from shortest wavelength to longest wavelength.

ANSWER: ____________________________

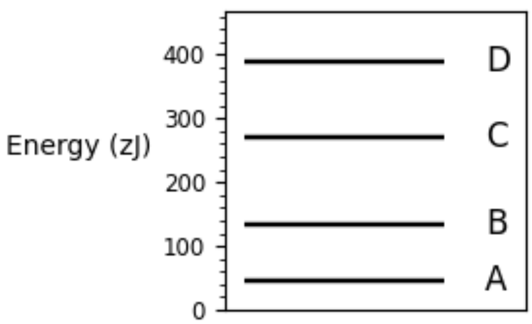
19. Suppose that there is heat transfer of 38 J to a system, while the system does 32 J of work. Later, there is heat transfer of 31 J out of the system while 31 J of work is done on the system. What is the net change in internal energy of the system?

ANSWER: ____________________________

20. An atom of a particular element has a radius of 189 pm and the average orbital speed of the electrons in it is approximately 1.34×10^7 m/s.
Calculate the least possible uncertainty in a measurement of the speed of an electron in this atom. Write your answer as a percentage of the average speed and round it to two significant figures.

ANSWER: ____________________________

21. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the longest wavelength?



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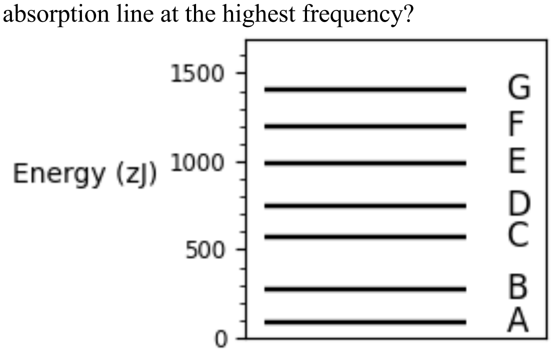
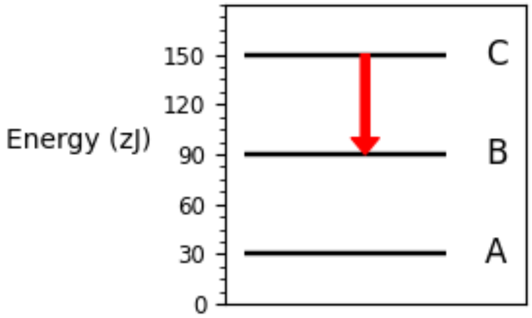
22. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the highest frequency?



ANSWER: ____________________________

23. This energy diagram shows the allowed energy levels of an electron in a certain atom. Suppose the electron makes a transition from C to B, as shown by the arrow. Calculate the wavelength of the photon that would be absorbed or emitted. Report your answer to three significant figures. Note that the SI prefix zepto (z) means 10^{-21}.

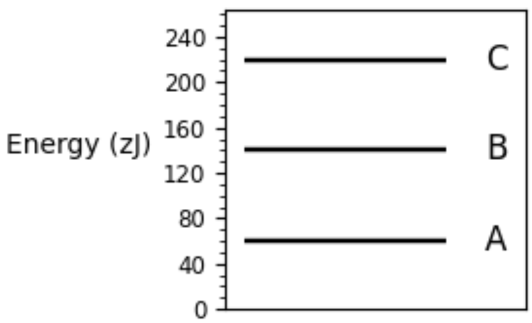


ANSWER: ______________________________

24. Imagine an alternate universe in which the value of the Planck constant is 3.44×10^{-32} J s. In that universe, is quantum mechanics necessary to accurately describe a 1.1×10^{-11}-m-wide object that has a mass of 1.3×10^{-25} kg and is moving at 1.4×10^{-6} m/s? Write "yes" if quantum mechanics is necessary and "no" if it is not, then explain why you made your choice. Answers lacking supporting explanation will earn no credit.

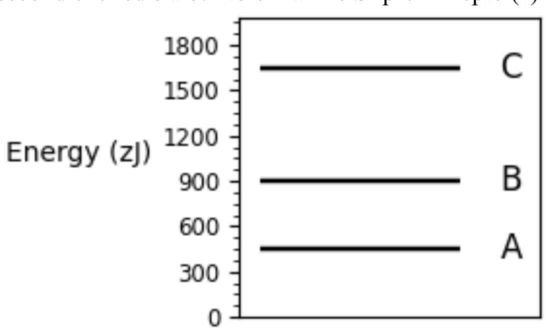
ANSWER: ______________________________

25. This energy diagram shows the allowed energy levels of an electron in a certain atom. What is the energy of the electron in its first excited state? Note that the SI prefix zepto (z) means 10^{-21}.



ANSWER: ______________________________

26. This energy diagram shows the allowed energy levels of an electron in a certain atom. What is the energy of the electron in its second excited state? Note that the SI prefix zepto (z) means 10^{-21}.



ANSWER: ______________________________

27. A chemist carefully measures the amount of heat needed to raise the temperature of a 1.4-kg sample of a pure substance from 6.7°C to 28.8°C. The experiment shows that 1.2 kJ of heat is needed. What can the chemist report for the specific heat capacity of the substance?

ANSWER: ______________________________

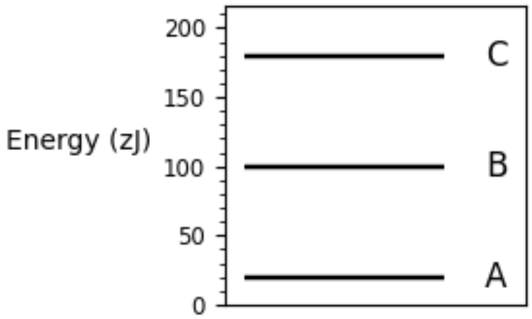
28. List the following electron subshells in order of increasing energy. You may assume that these subshells are in an atom with many electrons.
3p, 3s, 4d, 1s, 2s

ANSWER: ______________________________

29. Suppose a popular FM-radio station broadcasts radio waves with a frequency of 88.57 MHz. Calculate the wavelength of these radio waves.

ANSWER: ______________________________

30. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the shortest wavelength?

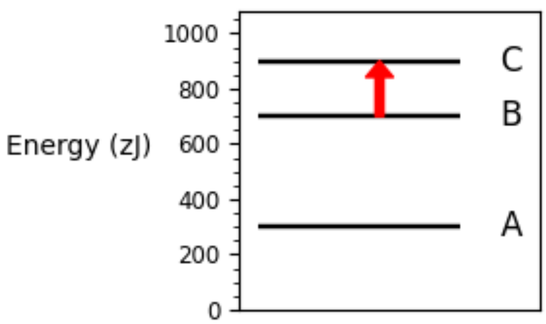


ANSWER: ______________________________

31. Michael puts a 0.318-kg aluminum pot on the stove and fills it with 0.456 liters of water. Both the pot and the water are initially at 0.566°C. How much heat is required to bring them both to 61.°C? The specific heat value of water is 4186. J/kg°C and for aluminum it is 900. J/kg°C.

ANSWER: ______________________________

32. This energy diagram shows the allowed energy levels of an electron in a certain atom. Suppose the electron makes a transition from B to C, as shown by the arrow. Calculate the energy of the photon that would be absorbed or emitted. Note that the SI prefix zepto (z) means 10^{-21}.



ANSWER: ______________________________